

NIKHIL DEEKONDA

5512295780 ♦ deekondanikhil100@gmail.com ♦ [Linkedin](#) ♦ [Github](#) ♦ [Portfolio](#)

EDUCATION

Yeshiva University, Katz School of Health and Science, NY, USA

Sep 2023 - May 2025

Masters of Science in Artificial Intelligence

CGPA 3.9

WORK EXPERIENCE

Valuai.io, Los Angeles, California, USA

May 2025 - Present

Lead AI Engineer

- Architected and deployed an AI-driven patient engagement platform integrating AWS Bedrock, LangChain, Redis, and MongoDB for secure, scalable clinician-patient interactions.
- Built multi-agent orchestration workflows with retrieval-augmented generation (RAG) to deliver personalized care recommendations, leveraging medical literature for evidence-based outputs.
- Fine-tuned large language models (DeepSeek-R1) to align with clinical guidelines, enabling context-aware and safe clinician-patient conversations that reduced interaction time.
- Developed and maintained a FastAPI backend, containerized with Docker, and deployed on AWS EC2 for scalability, supporting secure authentication (AWS Cognito), real-time text and voice interactions, and efficient Redis-based retrieval of patient and clinician-specific data.
- Deployed and served fine-tuned models on RunPod and FAL for low-latency inference, ensuring cost-efficient scaling for production workloads.
- Designed and launched a SwiftUI iOS application for secure, conversational patient engagement via text and voice, tightly integrated with backend AI services.
- Implemented dynamic prompt engineering for multi-turn agent interactions, optimizing responses based on personalized patient histories and clinical context.
- Built real-time monitoring and alert systems to notify clinicians of critical patient events, enabling proactive interventions and improving patient outcomes.

LTIMindtree, Pune, Maharashtra, India

Jul 2022 - Jul 2023

Senior Software Engineer

- Developed guardrails, data pipelines, and validation flows for ML/DL models including LLMs; fine-tuned models like GPT, Falcon, LLaMA-2, and Mistral for real-world use cases (PII removal, tone classification, toxic word detection) using LoRA and QLoRA techniques.
- Built a scalable platform to fine-tune LLMs with multiple techniques and implemented output validation for LLM agents utilizing frameworks such as AutoGen and LangGraph.
- Designed and executed complex Retrieval-Augmented Generation (RAG) workflows integrating LLMs with vector databases; developed hallucination detection systems leveraging Small Language Models (SLMs) and BERT-based models.
- Trained a proprietary 350M parameter language model from scratch and deployed fine-tuning, RAG, and validation workflows on AWS using services like EC2, S3, SageMaker, and Bedrock

National Highway Authority of India, Tirupati, Andhra Pradesh, India

Jun 2021 - Aug 2021

Machine Learning Intern

- Applied various machine learning algorithms to NHAI road dataset of half a million instances, evaluating models based on R-squared values, successfully predicting road rutting factors, based on the best performing model.
- Conducted model evaluation and feature analysis to identify key predictors, enabling data-driven insights for road quality and maintenance planning.

PROJECTS

TaskFin: RAG-Based Financial Task Automation

- Designed and implemented a RAG-based system for end-to-end bill payment automation, leveraging LangChain agents and the LangGraph framework to orchestrate multi-turn interactions and ensure context-aware, accurate financial task management from natural language input.
- Developed a robust Orchestrator Agent to coordinate specialized sub-agents for Authentication, Financial Transactions, and State Management, using LangChain memory (buffer, entity, summary) and MCP servers to maintain persistent contextual state across interactions.
- Designed detailed PRDs (Product Requirement Documents) before initiating development, ensuring clear objectives and measurable outcomes for the project.
- Deployed the complete solution as a fully interactive Streamlit conversational app, enabling real-world RAG-driven task automation, secure data handling, and rapid prototyping of AI-powered fintech assistants.

LungAware: AI Lung Cancer Detection App

- Engineered a deep convolutional neural network (CNN) for early lung cancer detection, achieving 99.4% classification accuracy through advanced image preprocessing, transfer learning, and hyperparameter optimization.
- Developed cross-platform (Android/iOS) mobile applications with Swift (iOS) and Kotlin (Android), integrating TFLite-quantized models and Grad-CAM for explainable AI and transparent diagnostic support.
- Implemented a seamless user interface and cloud-based backend (AWS EC2) for automated inference, patient management, and result notification, ensuring compliance with healthcare privacy and security best practices.

Enhancing LLM Reliability: Automated Fact-Checking with Cross-Encoder Model

- Built and fine-tuned a Microsoft DeBERTa v3 large language model as a fact-checking cross-encoder, achieving an F1 score of 85% on the Facebook Fact Checking Dataset and significantly reducing hallucination in downstream NLP systems.
- Automated real-time fact-checking in conversational and document QA scenarios, increasing the credibility and accuracy of LLM-generated content for end-users.

Optimizing Visual Bird Sound Denoising with Deep Learning: A Segmentation Approach

- Designed a custom encoder-decoder deep neural network architecture for visual bird sound denoising, outperforming state-of-the-art models with an IoU of 66.24 on challenging real-world datasets.
- Integrated advanced image segmentation and noise reduction algorithms, demonstrating measurable improvements in ecological acoustic monitoring and bioacoustics research.

SKILLS

Programming Languages	Python, Swift, Java, JavaScript, R, C, C++
Frameworks / Libraries	PyTorch, TensorFlow, FastAPI, LangChain, CrewAI, LangGraph, CoreML
Databases	PostgreSQL, MySQL, MongoDB, Redis
Cloud / DevOps	AWS, GCP, Docker, Git, GitHub
Tools / Analytics	Excel, Tableau, Jupyter, VS Code, XCode
Other	LLM Fine-tuning, RAG Pipelines, Multi-agent Systems, iOS Development (SwiftUI)

CERTIFICATIONS

- [AWS Certified Machine Learning Engineer – Associate](#)
- [AWS Certified AI Practitioner](#)
- [Engineer Data in Google Cloud](#)
- [Exploring Data With Looker](#)